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Group:

Computer Engineering Technology

Semester:

4th Semester

Subject:

Database Applications

Lab Task:

MySQL Lab Guide

Submitted To:

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Lab Task 5

Lab Tasks

1. Create the above tables with constraints.

The screenshot shows the phpMyAdmin web interface. The top navigation bar indicates the current context: Server: 127.0.0.1:3307, Database: college_db, and Table: employees. The left sidebar displays a tree view of the database structure, including 'college_db' and its tables 'courses' and 'employees'. The main area is the SQL query editor, which contains the following SQL statement:

```
1 CREATE TABLE departments (  
2   dept_id INT PRIMARY KEY,  
3   dept_name VARCHAR(100) UNIQUE  
4 );
```

Below the query editor, there are buttons for 'SELECT *', 'SELECT', 'INSERT', 'UPDATE', 'DELETE', 'Clear', 'Format', and 'Get auto-saved query'. A checkbox for 'Bind parameters' is also present. At the bottom, there is a 'Delimiter' dropdown set to ';', and several checkboxes: 'Show this query here again', 'Retain query box', 'Rollback when finished', and 'Enable foreign key checks' (which is checked). A 'Go' button is located to the right of these options. The bottom status bar shows 'Console'.

phpMyAdmin

Recent Favorites

Server: 127.0.0.1:3307 » Database: college_db » Table: employees

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Show query box

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0207 seconds.)

```
CREATE TABLE departments ( dept_id INT PRIMARY KEY, dept_name VARCHAR(100) UNIQUE );
```

[Edit inline] [Edit] [Create PHP code]

Console

phpMyAdmin

Recent Favorites

Server: 127.0.0.1:3307 » Database: college_db » Table: employees

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Run SQL query/queries on table college_db.employees:

```
1 CREATE TABLE students (  
2   student_id INT AUTO_INCREMENT PRIMARY KEY,  
3   name VARCHAR(100) NOT NULL,  
4   email VARCHAR(100) UNIQUE,  
5   age INT,  
6   dept_id INT,  
7   FOREIGN KEY (dept_id) REFERENCES departments(dept_id)  
8 );
```

emp_id
emp_name
salary
department
join_date
email
phone

phpMyAdmin

Recent Favorites

- New
- collegedb
- college_db
 - New
 - courses
 - departments
 - employees
 - students
- information_schema
- mysql
- performance_schema
- phpmyadmin

Server: 127.0.0.1:3307 » Database: college_db » Table: employees

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Show query box

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0242 seconds.)

```
CREATE TABLE students ( student_id INT AUTO_INCREMENT PRIMARY KEY, name VARCHAR(100) NOT NULL, email VARCHAR(100) UNIQUE, age INT, dept_id INT, FOREIGN KEY (dept_id) REFERENCES departments(dept_id) );
```

[Edit inline] [Edit] [Create PHP code]

Console

phpMyAdmin

Recent Favorites

- New
- collegedb
- college_db
 - New
 - courses
 - departments
 - employees
 - students
- information_schema
- mysql
- performance_schema
- phpmyadmin

Server: 127.0.0.1:3307 » Database: college_db » Table: employees

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Run SQL query/queries on table college_db.employees:

```
1 CREATE TABLE courses (  
2   course_id INT PRIMARY KEY,  
3   course_name VARCHAR(100),  
4   dept_id INT,  
5   FOREIGN KEY (dept_id) REFERENCES departments(dept_id)  
6 );  
7
```

emp_id
emp_name
salary
department
join_date
email
phone

SELECT * SELECT INSERT UPDATE DELETE Clear Format Get auto-saved query <<

☐ Bind parameters

Delimiter ; ☐ Show this query here again ☐ Retain query box ☐ Rollback when finished ☒ Enable foreign key checks Go

Console

phpMyAdmin

Recent Favorites

- New
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 - New
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 - employees
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 - students
- information_schema
- mysql
- performance_schema
- phpmyadmin

Server: 127.0.0.1:3307 » Database: college_db » Table: instructors

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0022 seconds.)

`SELECT * FROM `instructors``

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

instructor_id	name	email	dept_id
---------------	------	-------	---------

Query results operations

[Create view](#)

Console

phpMyAdmin

Recent Favorites

- New
- collegedb
- college_db
 - New
 - courses
 - departments
 - employees
 - students
- information_schema
- mysql
- performance_schema
- phpmyadmin

Server: 127.0.0.1:3307 » Database: college_db » Table: courses

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Run SQL query/queries on table college_db.courses:

```
1 CREATE TABLE instructors (  
2   instructor_id INT PRIMARY KEY,  
3   name VARCHAR(100),  
4   email VARCHAR(100) UNIQUE,  
5   dept_id INT,  
6   FOREIGN KEY (dept_id) REFERENCES departments(dept_id)  
7 );
```

course_id
course_name
dept_id

☐ Bind parameters

SELECT * SELECT INSERT UPDATE DELETE Clear Format Get auto-saved query <<

Delimiter ; ☐ Show this query here again ☐ Retain query box ☐ Rollback when finished ☒ Enable foreign key checks Go

Console

phpMyAdmin

Recent Favorites

- New
- collegedb
- college_db
 - New
 - courses
 - departments
 - employees
 - instructors
 - students
- information_schema
- mysql
- performance_schema
- phpmyadmin

Server: 127.0.0.1:3307 » Database: college_db » Table: instructors

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0022 seconds.)

```
SELECT * FROM `instructors`
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

instructor_id	name	email	dept_id
---------------	------	-------	---------

Query results operations

Create view

Console

phpMyAdmin

Recent Favorites

- New
- collegedb
- college_db
 - New
 - courses
 - departments
 - employees
 - instructors
 - students
- information_schema
- mysql
- performance_schema
- phpmyadmin

Server: 127.0.0.1:3307 » Database: college_db » Table: instructors

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Run SQL query/queries on table college_db.instructors:

```
1 CREATE TABLE enrollments (  
2   student_id INT,  
3   course_id INT,  
4   semester VARCHAR(20),  
5   PRIMARY KEY (student_id, course_id),  
6   FOREIGN KEY (student_id) REFERENCES students(student_id),  
7   FOREIGN KEY (course_id) REFERENCES courses(course_id)  
8 );
```

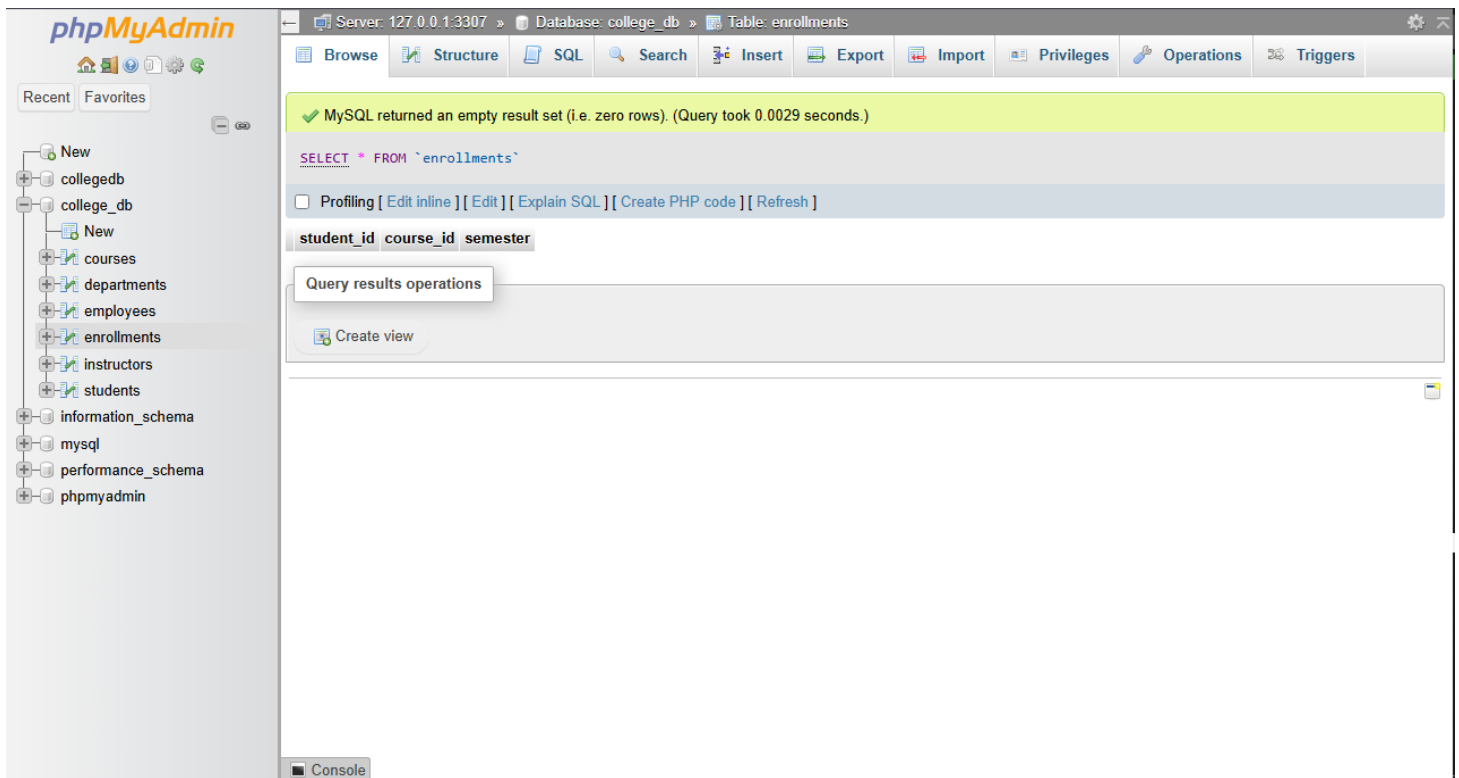
instructor_id
name
email
dept_id

SELECT * SELECT INSERT UPDATE DELETE Clear Format Get auto-saved query

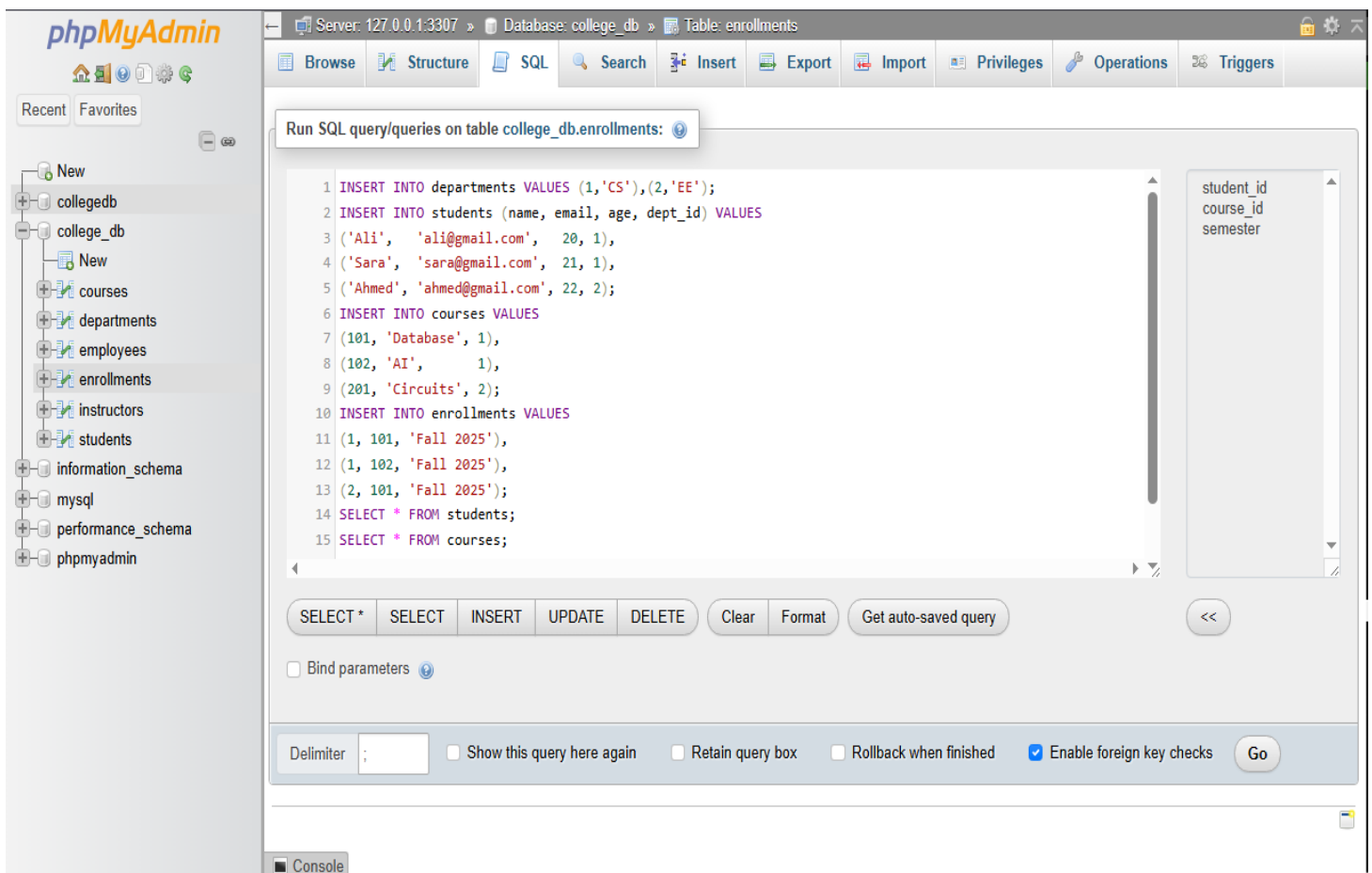
☐ Bind parameters

Delimiter ; ☐ Show this query here again ☐ Retain query box ☐ Rollback when finished ☒ Enable foreign key checks Go

Console



2.Insert data and query using SELECT.



phpMyAdmin

Server: 127.0.0.1:3307 » Database: college_db » Table: departments

Recent Favorites

New

- collegedb
- college_db
 - New
 - courses
 - departments
 - employees
 - enrollments
 - instructors
 - students
- information_schema
- mysql
- performance_schema
- phpmyadmin

Showing rows 0 - 1 (2 total, Query took 0.0006 seconds.)

```
SELECT * FROM `departments`
```

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	dept_id	dept_name
☐ Edit Copy Delete	1	CS
☐ Edit Copy Delete	2	EE

☐ Check all | With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

[Print](#) [Copy to clipboard](#) [Export](#) [Display chart](#) [Create view](#)

localhost/phpmyadmin/index.php?route=/database/structure&db=phpmyadmin

phpMyAdmin

Server: 127.0.0.1:3307 » Database: college_db » Table: students

Recent Favorites

New

- collegedb
- college_db
 - New
 - courses
 - departments
 - employees
 - enrollments
 - instructors
 - students
- information_schema
- mysql
- performance_schema
- phpmyadmin

Showing rows 0 - 2 (3 total, Query took 0.0031 seconds.)

```
SELECT * FROM `students`
```

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	student_id	name	email	age	dept_id
☐ Edit Copy Delete	1	Ali	ali@gmail.com	20	1
☐ Edit Copy Delete	2	Sara	sara@gmail.com	21	1
☐ Edit Copy Delete	3	Ahmed	ahmed@gmail.com	22	2

☐ Check all | With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

[Print](#) [Copy to clipboard](#) [Export](#) [Display chart](#) [Create view](#)

Console

phpMyAdmin

Recent Favorites

Server: 127.0.0.1:3307 » Database: college_db » Table: courses

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Showing rows 0 - 2 (3 total, Query took 0.0007 seconds.)

```
SELECT * FROM `courses`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

		course_id	course_name	dept_id
☐	Edit Copy Delete	101	Database	1
☐	Edit Copy Delete	102	AI	1
☐	Edit Copy Delete	201	Circuits	2

Check all | With selected: Edit Copy Delete Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Print Copy to clipboard Export Display chart Create view

Console

phpMyAdmin

Recent Favorites

Server: 127.0.0.1:3307 » Database: college_db » Table: enrollments

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Showing rows 0 - 2 (3 total, Query took 0.0008 seconds.)

```
SELECT * FROM `enrollments`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

		student_id	course_id	semester
☐	Edit Copy Delete	1	101	Fall 2025
☐	Edit Copy Delete	1	102	Fall 2025
☐	Edit Copy Delete	2	101	Fall 2025

Check all | With selected: Edit Copy Delete Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Print Copy to clipboard Export Display chart Create view

Console

3. Update a student's name.

The image displays two screenshots of the phpMyAdmin interface, demonstrating the process of updating a student's name in a database.

Top Screenshot: SQL Query Execution

The interface shows the 'SQL' tab selected. The query editor contains the following SQL code:

```
1 UPDATE students
2 SET name = 'Ali Khan'
3 WHERE student_id = 1;
4 SELECT * FROM students WHERE student_id = 1;
```

The table structure for 'students' is visible on the right:

student_id
name
email
age
dept_id

Below the query editor, the 'Simulate query' button is highlighted. The 'Delimiters' section shows the delimiter as semicolon (;). The 'Bind parameters' checkbox is unchecked. The 'Show this query here again' checkbox is unchecked. The 'Retain query box' checkbox is unchecked. The 'Rollback when finished' checkbox is unchecked. The 'Enable foreign key checks' checkbox is checked.

Bottom Screenshot: Query Results

The interface shows the 'Query results' tab selected. The query executed was:

```
SELECT * FROM `students`
```

The results show 3 rows (total 3 rows, Query took 0.0009 seconds):

student_id	name	email	age	dept_id
1	Ali Khan	ali@gmail.com	20	1
2	Sara	sara@gmail.com	21	1
3	Ahmed	ahmed@gmail.com	22	2

The 'Query results operations' section shows buttons for 'Print', 'Copy to clipboard', 'Export', 'Display chart', and 'Create view'.

4.Delete a student record.

The screenshot shows the phpMyAdmin interface with the 'enrollments' table selected. The SQL query editor contains the following queries:

```
1 DELETE FROM enrollments WHERE student_id = 2;  
2 DELETE FROM students WHERE student_id = 2;  
3 SELECT * FROM students;
```

The interface includes a sidebar with a database tree, a top navigation bar with tabs like 'Browse', 'Structure', 'SQL', 'Search', 'Insert', 'Export', 'Import', 'Privileges', 'Operations', and 'Triggers'. Below the query editor, there are buttons for 'SELECT *', 'SELECT', 'INSERT', 'UPDATE', 'DELETE', 'Clear', 'Format', and 'Get auto-saved query'. There is also a checkbox for 'Bind parameters' and a section for query options including 'Delimiter', 'Show this query here again', 'Retain query box', 'Rollback when finished', and 'Enable foreign key checks' (checked). 'Simulate query' and 'Go' buttons are at the bottom.

This screenshot shows the same phpMyAdmin interface as the previous one, but with a confirmation dialog box overlaid. The dialog box is titled 'localhost says' and contains the text: 'Do you really want to execute "DELETE FROM enrollments WHERE student_id = 2; DELETE FROM students WHERE student_id = 2; SELECT * FROM ...?"'. It has 'OK' and 'Cancel' buttons.

The screenshot shows the phpMyAdmin interface after the query has been executed successfully. A green message bar at the top states: 'Showing rows 0 - 1 (2 total, Query took 0.0006 seconds.)'. Below this, the query 'SELECT * FROM `enrollments`' is displayed. At the bottom, there are links for 'Profiling', 'Edit inline', 'Edit', 'Explain SQL', 'Create PHP code', and 'Refresh'.

phpMyAdmin

Recent Favorites

New

- collegedb
- college_db
 - New
 - courses
 - departments
 - employees
 - enrollments
 - instructors
 - students
- information_schema
- mysql
- performance_schema
- phpmyadmin

Server: 127.0.0.1:3307 » Database: college_db » Table: students

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Showing rows 0 - 1 (2 total, Query took 0.0010 seconds.)

SELECT * FROM `students`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	student_id	name	email	age	dept_id
☐ Edit Copy Delete	1	Ali Khan	ali@gmail.com	20	1
☐ Edit Copy Delete	3	Ahmed	ahmed@gmail.com	22	2

Check all | With selected: Edit Copy Delete Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Print Copy to clipboard Export Display chart Create view

Console

5. Try TRUNCATE and DROP.

phpMyAdmin

Recent Favorites

New

- collegedb

Server: 127.0.0.1:3307 » Database: college_db » Table: students

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Run SQL query/queries on table college_db.students:

```
1 TRUNCATE TABLE enrollments;
2 TRUNCATE TABLE students;
```

student_id
name

localhost/phpmyadmin/index.php?route=/table/sql&db=college_...

voice maker Love Calculator | Lo... Like4Like -

phpMyAdmin

Recent Favorites

New

collegedb

college_db

New

courses

departments

employees

enrollments

instructors

students

information_schema

mysql

performance_schema

phpmyadmin

Server: 127.0.0.1:3306

Browse Structure

Run SQL query/queries

1 TRUNCATE TABLE

2 TRUNCATE TABLE

3

4 DROP TABLE enr

5 DROP TABLE stu

6 DROP TABLE courses;

7 DROP TABLE instructors;

8 DROP TABLE departments;

SELECT *

SELECT

INSERT

UPDATE

DELETE

Clear

Format

Get auto-saved query

<<

☐ Bind parameters

Delimiter ;

☐ Show this query here again

☐ Retain query box

☐ Rollback when finished

☒ Enable foreign key checks

Go

localhost says

Do you really want to execute "TRUNCATE TABLE enrollments;
TRUNCATE TABLE students;

DROP TABLE enrollments;
DROP TABLE students;
D
...?"

OK Cancel

Privileges Operations Triggers

student_id
name
email
age
dept_id

phpMyAdmin

Recent Favorites

New

collegedb

college_db

New

employees

information_schema

Server: 127.0.0.1:3307 » Database: college_db » Table: students

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Show query box

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0277 seconds.)

DROP TABLE enrollments;

[Edit inline] [Edit] [Create PHP code]

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0070 seconds.)

6. Add a new column using ALTER TABLE.

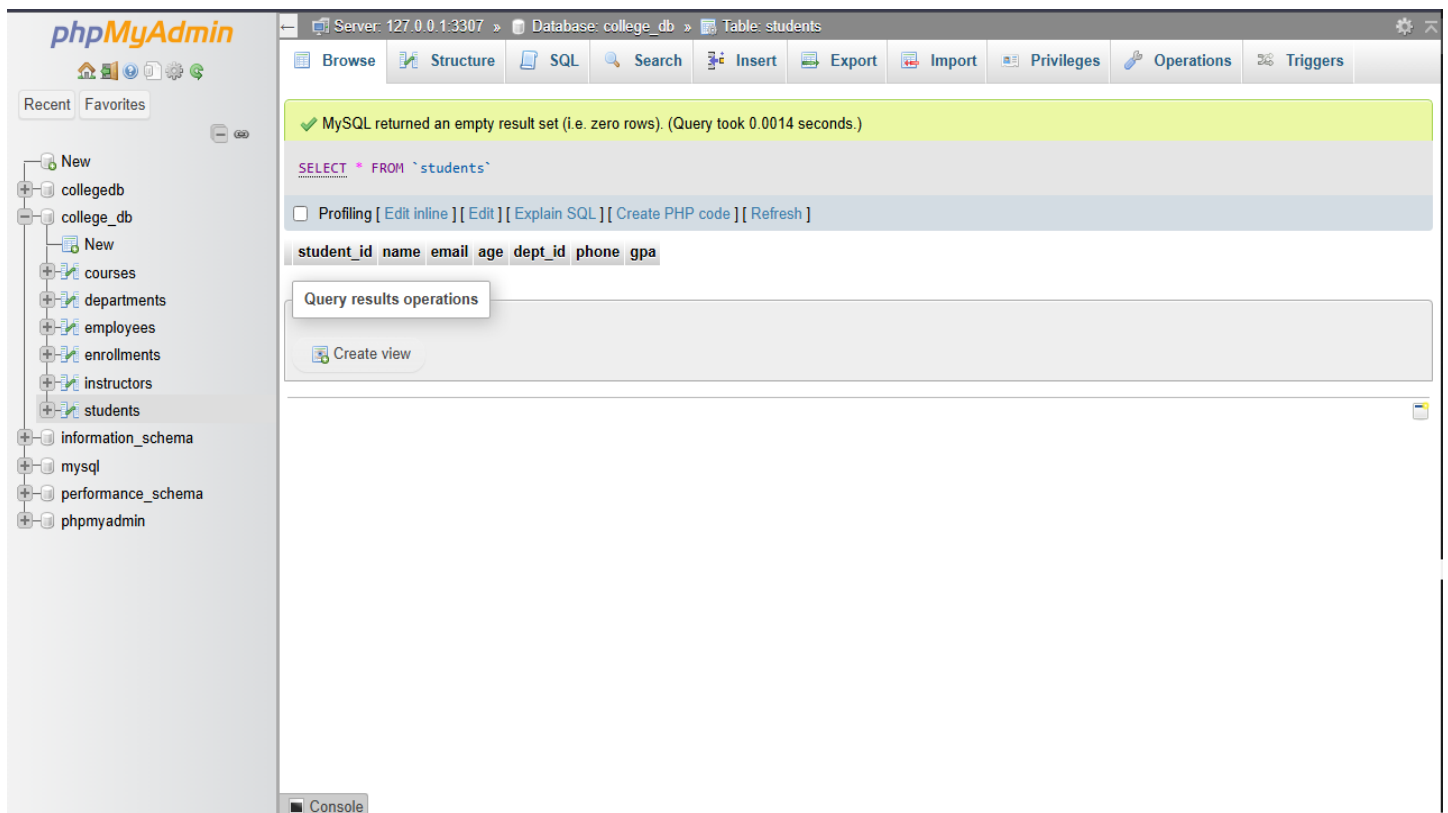
The screenshot shows the phpMyAdmin interface with the 'students' table selected in the 'college_db' database. The SQL query editor contains the following commands:

```
1 ALTER TABLE students
2 ADD phone VARCHAR(20);
3 ALTER TABLE students
4 ADD gpa DECIMAL(3,2) DEFAULT 0.00;
5
6 DESCRIBE students;
```

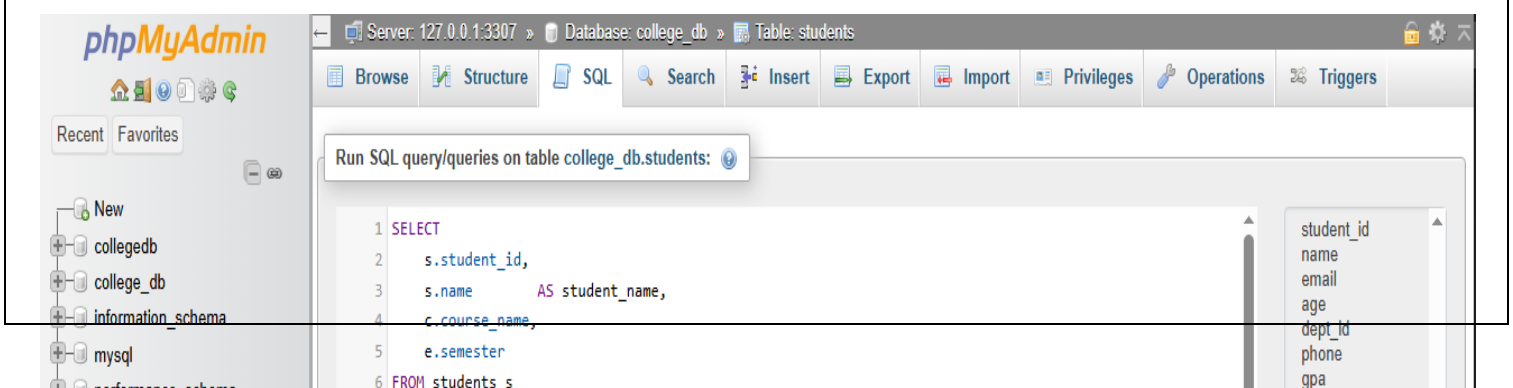
The table structure on the right shows the existing columns: student_id, name, email, age, and dept_id. Below the query editor, there are buttons for 'SELECT *', 'SELECT', 'INSERT', 'UPDATE', 'DELETE', 'Clear', 'Format', and 'Get auto-saved query'. There is also a checkbox for 'Bind parameters' and a 'Go' button at the bottom right.

The screenshot shows the phpMyAdmin interface after executing the SQL queries. The 'Show query box' button is visible. The execution results are displayed in a list:

- ✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0293 seconds.)
`ALTER TABLE students ADD phone VARCHAR(20);`
[Edit inline] [Edit] [Create PHP code]
- ✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0074 seconds.)
`ALTER TABLE students ADD gpa DECIMAL(3,2) DEFAULT 0.00;`



7.Practice joins between Students and Courses.



8. Explore candidate keys, alternate keys, and composite keys.

Composite Key Example (already in enrollments)

student_id + course_id together = one unique record

PRIMARY KEY (student_id, course_id)

Candidate Keys in students table:

Both student_id and email can uniquely identify a student

student_id → chosen as PRIMARY KEY

